

A Cyber Risk-Integrated Cost-Benefit Analysis Framework for HIPAA-Compliant Infrastructure Modernization in Hospitals



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Abstract

Aim: This study examines the economic and risk-management implications of hospital information technology (IT) infrastructure modernization within the regulatory context of the Health Insurance Portability and Accountability Act (HIPAA). The primary aim is to develop a cyber risk-integrated cost-benefit analysis framework tailored to HIPAA-regulated hospital environments in order to support risk-informed infrastructure investment decisions.

Methods: The study develops a structured cost-benefit analysis (CBA) model that incorporates cybersecurity risk quantification into traditional financial evaluation methodologies. Regulatory requirements enforced by the U.S. Department of Health and Human Services Office for Civil Rights are analyzed to define compliance-driven cost exposures. Cyber risk is operationalized through monetization of breach probability and impact using Expected Annualized Loss (EAL). The framework accounts for heterogeneous asset lifecycles, legacy system constraints, and probabilistic threat exposure. A realistic hospital-based case illustration is employed to demonstrate practical applicability of the proposed model.

Results: The analysis indicates that legacy hospital infrastructures significantly elevate risks of downtime, data breaches, and patient safety incidents, particularly when outdated operating systems and unsupported medical devices cannot accommodate modern security controls. Breach remediation costs and regulatory penalties substantially increase total exposure in non-modernized environments. By integrating EAL into cost-benefit modeling, the framework captures both direct financial expenditures and risk-adjusted benefits associated with cybersecurity enhancements. The results demonstrate that modernization initiatives, including secure architecture redesign and improved system resilience, generate measurable long-term economic value when evaluated over extended time horizons.

Conclusion: Enterprise-wide IT infrastructure modernization in hospitals must be evaluated not only as a capital expenditure decision but also as a risk mitigation investment under HIPAA compliance constraints.

Recommendation: Hospitals should adopt risk-adjusted economic evaluation models that integrate cybersecurity exposure metrics, including Expected Annualized Loss, into infrastructure investment planning.

Keywords: *HIPAA, cost-benefit analysis, cyber risk, zero trust architecture, healthcare IT, cybersecurity economics*

1. INTRODUCTION

Information technology infrastructure modernization has become a strategic imperative for hospitals due to increasing digitization of clinical workflows, rising cybersecurity threats, and intensifying regulatory oversight. Modern healthcare delivery relies on interoperable electronic health record (EHR) systems, networked medical devices, and data-driven decision support tools, all of which depend on resilient and secure IT infrastructure. When infrastructure is misaligned with organizational strategy or constrained by legacy technologies, hospitals face elevated risks of system downtime, data breaches, and compromised patient safety.

Under HIPAA, covered entities are required to implement administrative, physical, and technical safeguards to ensure the confidentiality, integrity, and availability of ePHI. Enforcement actions by the U.S. Department of Health and Human Services (HHS) Office for Civil Rights demonstrate that non-compliance can result in substantial financial penalties and reputational harm. In parallel, healthcare has emerged as a prime target for ransomware and supply-chain attacks, with breach remediation costs significantly exceeding those observed in other sectors.

Legacy medical devices and outdated operating systems present particular challenges. Many clinical systems operate on unsupported platforms that cannot accommodate modern security controls, such as endpoint protection, encryption, or network segmentation. These constraints increase vulnerability to cyber incidents and complicate compliance efforts. Despite these risks, hospitals often hesitate to invest in comprehensive modernization due to high upfront capital costs and uncertainty regarding financial returns.

Traditional economic evaluations in healthcare have focused primarily on discrete technologies such as EHR implementations. However, full-scale infrastructure modernization under HIPAA constraints introduces additional complexity, including heterogeneous asset lifecycles and probabilistic cybersecurity risks. These characteristics necessitate a specialized CBA framework capable of capturing both tangible and intangible benefits over extended time horizons.

This study makes three primary contributions. First, it extends traditional hospital cost–benefit analysis by explicitly integrating cybersecurity risk and regulatory exposure into economic evaluation. Second, it proposes a structured and reproducible workflow that monetizes breach risk using expected annualized loss (EAL) within a HIPAA-compliant context. Third, it demonstrates the practical applicability of the framework through a realistic hospital-based case illustration, supporting informed infrastructure investment decisions under uncertainty.

1.1 Research Gap

While prior studies have examined cost–benefit and return-on-investment analyses for electronic health records (EHRs) and digital health applications, limited research has extended these methods to enterprise-wide IT infrastructure modernization under regulatory and cybersecurity constraints. Systematic reviews of digital health economic evaluations indicate that cybersecurity risk reduction, regulatory exposure, and infrastructure resilience are frequently under-quantified or treated as external considerations rather than economic variables [1], [6]. Existing hospital CBA and ROI frameworks emphasize productivity gains, clinical outcomes, and organizational efficiency but rarely incorporate probabilistic cyber loss or HIPAA enforcement risk within discounted cash-flow analysis [2], [3]. As a result, healthcare decision-makers lack robust economic frameworks that adequately capture the financial implications of cyber risk and compliance-driven modernization initiatives. This gap is

particularly significant in HIPAA-regulated hospital environments, where breach costs and regulatory penalties can materially affect long-term financial performance [7], [11].

1.2 Objective of this paper:

The objectives of this paper are to: (i) design a cyber risk–integrated cost–benefit analysis framework for HIPAA-compliant hospital IT infrastructure modernization; (ii) operationalize cybersecurity risk as a monetizable economic variable within the CBA process; and (iii) demonstrate how the proposed framework supports long-term, risk-informed infrastructure investment decisions in healthcare organizations.

2. LITERATURE REVIEW

Economic evaluation of digital health interventions presents methodological challenges due to rapid technological evolution, system interdependencies, and non-clinical outcome measures. Prior research emphasizes the need to extend traditional health economic methods beyond direct clinical outcomes to capture organizational, infrastructural, and risk-related impacts [1], [3]. Frameworks such as eHealth-CBA highlight the importance of aggregating direct costs, productivity effects, and phased benefits but provide limited guidance on quantifying cybersecurity or regulatory exposure [1].

Systematic reviews of healthcare return-on-investment models further demonstrate that intangible benefits such as quality improvement, staff satisfaction, and organizational learning play a significant role in value realization but are rarely monetized consistently across institutions [3][4]. Studies examining Healthcare 4.0 and digitally enabled hospital environments similarly report gaps in the costing of integration, maintenance, and cybersecurity controls, limiting the transferability of findings across healthcare systems [5], [16]. Recent reviews of AI-enabled decision support and advanced analytics systems show a growing body of health economic evaluations; however, many studies lack a comprehensive accounting for deployment, maintenance, interoperability, and security-related costs [6]. These omissions are particularly problematic in hospital environments, where cybersecurity incidents and regulatory non-compliance can result in substantial financial losses.

A consistent theme across the literature is the underrepresentation of cybersecurity and regulatory risk in economic evaluations of healthcare IT investments. Economic models from the information security domain demonstrate that organizations systematically underinvest in cybersecurity when risk is not explicitly monetized [7][8][9]. In contrast to existing healthcare-focused frameworks, this study positions cybersecurity risk and HIPAA compliance as central economic variables rather than external constraints, addressing a critical gap in the evaluation of hospital IT infrastructure modernization.

3. METHODOLOGY

The proposed framework applies a discounted cash-flow cost–benefit analysis over a multi-year horizon (5–15 years) to reflect the long-lived nature of hospital IT infrastructure assets. Costs are categorized into capital expenditures (CapEx), operational expenditures (OpEx), compliance and security implementation costs, workforce training, and transition-related expenses. Benefits include operational efficiency gains, clinical quality improvements, compliance risk reduction, and long-term strategic flexibility. Net present value (NPV), benefit–cost ratio (BCR), and internal rate of return (IRR) are used as primary decision metrics, consistent with established healthcare economic evaluation practices [1], [2].

Cybersecurity risk is operationalized using expected annualized loss (EAL), defined as the product of breach probability and estimated breach impact. This approach is grounded in

information security economics literature, which demonstrates the importance of probabilistic loss modelling for rational security investment decisions [8], [19].

Sensitivity and scenario analyses are incorporated to address uncertainty in breach probabilities, discount rates, operational savings, and adoption outcomes. Best-case, moderate-case, and worst-case scenarios are evaluated to test model robustness, aligning with recommended practices in both healthcare IT and cybersecurity investment evaluation [7], [8].

3.1 Empirical Evidence on Financial Impacts

EHR/EMR investments: Several recent empirical reviews and primary studies report operational gains (reduced storage, faster information retrieval) and improvements in process metrics; however, measurable financial returns vary substantially across settings and implementation quality. Low et al. (2013) and more recent 2024–2025 studies show positive clinical/process signals, but financial payback is institution-specific and sensitive to assumptions about labour savings and reuse of clinician time.

H4.0 / Industry 4.0 hospital technologies: Reviews of hospital adoption of H4.0 technologies find that standard capital-investment evaluation techniques (discounted cash flow, CBA) are widely used, but often without adequately modelling interoperability, device heterogeneity, and security externalities. Vassolo et al. (2021) recommend combining engineering and economic modelling for such investments.

AI and decision support economic evaluations: Systematic reviews (e.g., Voets et al., 2022) show an increasing but still nascent body of health economic evaluations for AI tools; many lack robust costing of deployment, maintenance, and integration, limiting transferability.

A recurring empirical theme: risk reduction (e.g., fewer breaches, lower remediation costs) and regulatory compliance are important but under-quantified in many studies — yet they are central when HIPAA liability is considered.

3.2 Cyber Risk-Integrated Cost-Benefit Framework

The proposed methodology applies a structured, multi-stage cost–benefit analysis to evaluate hospital IT infrastructure modernization under HIPAA and cybersecurity constraints. The framework follows five sequential steps: Baseline assessment, cost identification and categorization, cyber risk quantification, economic evaluation, sensitivity and scenario analysis.

Baseline assessment entailed identification of existing infrastructure assets, legacy system dependencies, historical downtime, security incidents, and HIPAA compliance posture. Cost identification and categorization involved classification of capital expenditures (CapEx), operational expenditures (OpEx), compliance and security implementation costs, workforce training, and transition-related expenses. Cyber risk quantification involved estimation of baseline and post-modernization cyber risk using expected annualized loss (EAL), calculated as the probability of a breach multiplied by its estimated financial impact. On the other hand, economic evaluation entailed application of discounted cash-flow techniques over a multi-year horizon (5–15 years) using net present value (NPV), benefit–cost ratio (BCR), and internal rate of return (IRR). Sensitivity and scenario analysis entailed evaluation of best-case, moderate-case, and worst-case scenarios by varying breach probabilities, discount rates, operational savings, and adoption effectiveness.

This structured approach enhances transparency, reproducibility, and decision relevance for hospital executives, compliance officers, and IT governance bodies.

4. CYBER RISK-INTEGRATED CBA WORKFLOW

4.1 Capital Expenditures

The capital expenditures or CapEx are the main cost category to be taken into consideration, because network hardware, servers, redundant systems, network segmentation equipment, zero trust architecture implementation and secure cloud infrastructure setup.

4.2 Compliance and Security Implementation

Compliance and security implementation includes risk assessments, gap analysis, policies and procedures documentation, encryption and decryption of software and necessary hardware, access control and identity management systems, audit logging/monitoring tools, and security operations (SOC) setup.

4.3 Training and Human Resources

Organizations must provide structured training programs for IT personnel, clinicians, and administrative staff covering cybersecurity awareness, HIPAA compliance, and incident response procedures. Regular refresher sessions, phishing awareness programs, and role-based security training ensure sustained compliance and operational resilience.

4.4 Maintenance and Operations

OpEx includes ongoing maintenance, software updates/patches, license renewals, auditing, monitoring, and compliance reassessments.

5. Benefit CATEGORIES

Here, we have considered various cost categories to get the maximum output with minimum input because every healthcare organization has certain budget limits to run the whole environment.

5.1 Clinical quality and safety

With this implementation in the health care industry, one can reduce the medical errors, medication errors and improve coordination across various departments by improving clinical decisions and improving care continuity, which helps the entire organization to achieve more success and deliver quality care and improved systems.

5.2 Long-term sustainability and future readiness

Another benefit includes long-term sustainability and future readiness for the entire health care system and improved health care because of the ability to support future digital health initiatives, which include Zero Trust Architecture implementation, artificial intelligence and machine learning systems by integrating IoTs and data analytics systems. Also, interoperability with the help of easier system integrations not only lowers the cost of upgrades but also increases the profit and overall system in the organization.

5.3 Compliance risk reduction

Compliance risk reduction is the main benefit among all the other forms, which helps in reducing the amount of data breaches, HIPAA violations, heavy fines and penalties and breach-related costs such as legal, remediation and reputational.

5.4 Operational efficiency improvement

Even with small error-rate reductions, they can translate into major savings such as labor productivity improvements, process automation and error reduction and system availability and

downtime reduction. This helps in efficiency outcome, shorter cycle times, faster decision-making and reduced administrative friction.

5.5 Social and human capital value

Better patient care, improved staff satisfaction, improved data quality and efficiency, improved institutional reputation, resilience against cyber threats, though harder to monetize, may be valued qualitatively as per general cost-benefit analysis theory.

6 ANALYTICAL/FINANCIAL MODEL

The economic evaluation applies a discounted cash flow approach over a multi-year horizon of 5-15 years to reflect the long-term nature of infrastructure investments.

6.1 Core Evaluation Metrics

Net Present Value (NPV)

$$NPV = \sum_{t=0}^T (B_t - C_t) / (1 + r)^t$$

where B_t represents benefits in year t , C_t represents costs in year t , and r is the discount rate.

Benefit-Cost Ratio (BCR)

$$BCR = \frac{\sum B_t}{\sum C_t}$$

Used as an intuitive indicator of economic efficiency.

Internal Rate of Return (IRR)

Defined as the discount rate at which the NPV equals zero, enabling comparison with alternative investment options.

6.2 Sensitivity and Uncertainty Analysis

Given the uncertainty inherent in estimating long-term operational and compliance benefits, sensitivity analyses are conducted across multiple scenarios. Key parameters that varied include discount rates, probability of security incidents, maintenance costs, staff adoption rates, and projected efficiency gains. Best-case, moderate-case, and worst-case scenarios are evaluated to test model robustness.

7. COST-BENEFIT ANALYSIS MODEL FOR HOSPITAL IT INFRASTRUCTURE MODERNIZATION

7.1 Cost Structure and estimated ranges for hospital IT infrastructure modernization

This study categorizes costs into capital, operational, human resource, compliance, and transition-related components. Cost estimates are expressed as ranges to account for variation in hospital size, system complexity, and vendor pricing.

Table 1: Cost Categories and Estimated Ranges

Cost Category	Description	Estimated Cost Range (USD)	Key Assumptions
Capital & Investment	Acquisition of servers, storage systems, networking hardware, segmentation controls, firewalls, and intrusion detection/prevention technologies	\$200,000–\$600,000 (one-time)	Scale depends on bed count and existing infrastructure maturity
Security/Compliance Implementation	Deployment of encryption mechanisms, identity and access controls, audit logging, risk assessments, and penetration testing	\$25,000–\$100,000 (initial) plus \$5,000–\$20,000 annually	Assumes a mid-sized hospital with baseline security controls
Training/Workforce Enablement	Initial staff training and periodic refresher programs covering system use, security awareness, and compliance procedures	\$5,000–\$20,000 per year	Cost varies with staff size and training modality
Operations/Maintenance (OpEx)	Ongoing licensing, software updates, monitoring, backup services, vendor support, and compliance audits	\$10,000–\$50,000 per year	Reflects hybrid on-prem/cloud operations
Data Migration Transition	Data conversion, parallel system operation, testing, validation, and productivity loss during transition	\$50,000–\$200,000 (one-time)	Strongly influenced by legacy system complexity

Note: Cost ranges reflect mid-sized hospitals (approximately 150–300 beds) and are derived from industry benchmarks, prior healthcare IT modernization studies, and vendor-reported implementation estimates. Actual costs may vary based on infrastructure maturity and deployment scope.

7.2 Benefit Identification and Valuation

Benefits are classified as direct operational gains, clinical performance improvements, compliance risk mitigation, long-term strategic value, and intangible organizational outcomes. Where direct monetization is not feasible, proxy indicators are applied.

Table 2: Benefit Identification and Valuation

Benefit Category	Measurement Approach	Estimated Annual Benefit (USD)	Valuation Basis
Operational Efficiency	Reduction in administrative workload, elimination of redundant processes, faster information retrieval	\$100,000–\$300,000	Monetized using staff time savings multiplied by the average labour cost
Clinical Quality and Safety	Lower error incidence, fewer adverse events, improved patient flow and throughput	\$50,000–\$200,000 plus non-monetary benefits	Requires comparison of pre- and post-implementation performance indicators
Compliance & Cyber Risk Reduction	Decreased likelihood of regulatory penalties, breach response costs, and insurance premiums	\$500,000–\$2,000,000 (risk-adjusted)	Estimated using probability-weighted breach avoidance
Scalability and Future Readiness	Deferred upgrade expenses, improved system interoperability, and reduced future reengineering	\$20,000–\$100,000	Calculated as avoided or postponed capital expenditures
Intangible and Social Value	Staff satisfaction, patient trust, and organizational resilience	Not directly monetized	Assessed through proxy measures such as turnover reduction

Cyber risk reduction benefits are estimated using probability-weighted avoided loss calculations based on reported healthcare breach costs and observed reductions in breach likelihood following implementation of network segmentation, identity and access management, endpoint modernization, and continuous monitoring. Average breach impact values are drawn from published healthcare cybersecurity reports, while probability reductions reflect conservative post-modernization assumptions reported in industry and regulatory analyses.

8. Cyber Risk–Integrated CBA Workflow

Figure 1 illustrates the proposed cyber risk–integrated cost–benefit analysis workflow, embedding cybersecurity risk quantification directly into the economic evaluation of hospital IT infrastructure modernization.

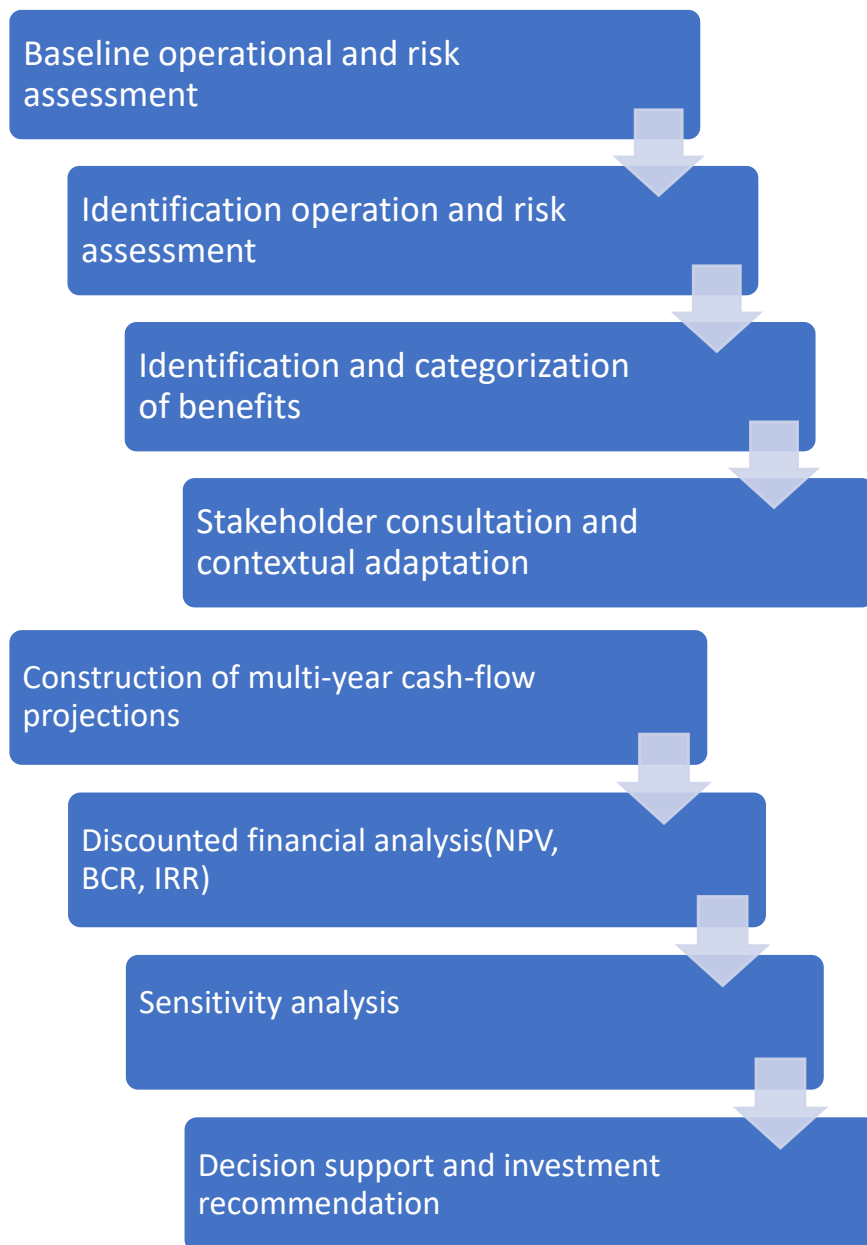
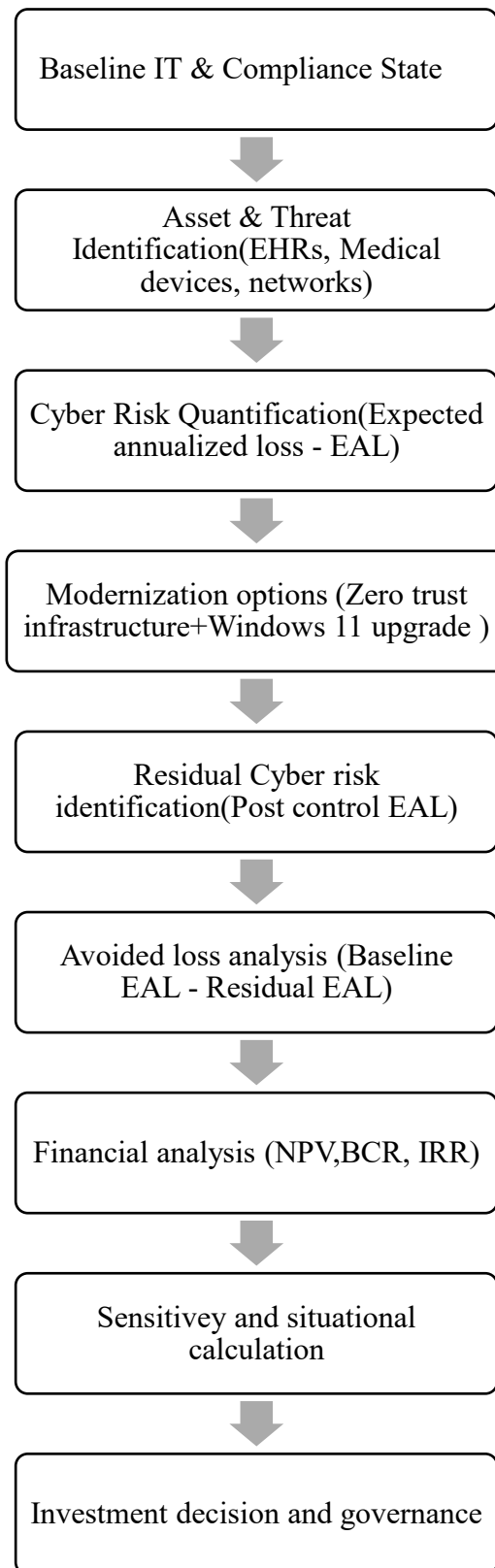


Figure 1: Cyber Risk-Integrated Workflow Illustrating how Modernization Costs, Cybersecurity Risk Quantification (EAL), HIPAA Compliance Requirements, and Financial Evaluation Metrics (NPV, IRR, BCR) Interact Within the Hospital Infrastructure Investment Decision Process

8.1 CBA Workflow with Cyber Risk Integration

This framework integrates cybersecurity risk directly into cost-benefit analysis, ensuring that HIPAA compliance and breach-risk reduction are treated as economic variables rather than externalities. The workflow aligns technical security controls with financial decision-making and governance.



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Figures 2: Integrated Workflow Demonstrating the Embedding of Cybersecurity Risk and HIPAA Compliance Considerations Into the Cost-Benefit Analysis Model for Hospital IT Infrastructure Modernization

Source: Author

9. Worked Mini-Example

To illustrate the application of the proposed framework, a mid-sized 200-bed hospital operating legacy clinical systems and a flat network architecture is considered. Healthcare organizations of this scale have been shown to face significant cybersecurity exposure due to device heterogeneity and limited segmentation capabilities [11], [12].

Based on sector benchmarks, the annual probability of a major data breach is estimated at 18%, with an average financial impact of approximately USD 7.5 million, reflecting incident response costs, downtime, legal exposure, and reputational loss [11]. Following infrastructure modernization, including Zero Trust Architecture implementation and operating system upgrades, the breach probability is conservatively reduced to 5%, consistent with risk-reduction estimates reported in cybersecurity economics literature [8].

Over a 10-year evaluation horizon and using a conservative 5% discount rate, the present value of avoided cyber losses alone exceeds USD 7.5 million. This substantially outweighs the initial modernization investment and recurring operational costs, demonstrating that cyber-risk-adjusted benefits can independently justify infrastructure modernization in HIPAA-regulated hospital environments [8], [19]

9.1 Baseline Cyber Risk

Based on sector benchmark, the annual probability of a major breach is estimated at 18%, with an average impact of USD 7.5 Million (Incident response, downtime, legal exposure, reputational loss).

Baseline EAL = $0.18 \times 7.5\text{M} = \text{USD } 1.35 \text{ million/year}$.

9.2 Modernization Investment

The hospital invests USD 750,000 in infrastructure modernization (ZTA infrastructure + Windows 11 upgrades), with annual operating costs of USD 120,000.

9.3 Post-Modernization Risk

Here, security controls reduce breach probability to 5%.

Residual EAL = $0.05 \times 7.5\text{M} = \text{USD } 375,000/\text{year}$.

9.4 Avoided Cyber loss

$\Delta\text{EAL} = 1.35\text{M} - 0.375\text{M} = \text{USD } 975,000/\text{year}$.

9.5 Economic Outcome:

Over a 10-year evaluation horizon and using a conservative 5% discount rate, the present value of avoided cyber losses alone exceeds USD 7.5 million. This substantially outweighs the initial modernization investment of USD 750,000 and recurring annual operating costs of approximately USD 120,000. Even when excluding operational efficiency gains, clinical quality improvements, and strategic benefits, the cyber-risk-adjusted benefits generate a positive net present value. When additional operational and compliance-related benefits are included, the overall economic justification for infrastructure modernization becomes significantly stronger, demonstrating the robustness of the proposed framework under conservative assumptions.

10. RESULTS AND DISCUSSION

10.1 Key Findings

The application of the cyber risk–integrated cost–benefit framework yields several important findings. First, cybersecurity risk reduction constitutes a dominant economic benefit in HIPAA-regulated hospital environments, often exceeding traditional operational efficiency gains. Second, infrastructure modernization initiatives demonstrate positive net present value even under conservative breach probability reduction assumptions, highlighting the financial materiality of cyber risk [7], [8].

Treating HIPAA compliance as an economic variable—rather than a regulatory afterthought—materially alters investment prioritization and strengthens governance accountability. These findings align with IT governance literature emphasizing the importance of risk-informed decision-making for capital allocation in complex organizations [18].

10.2 Policy Implications

From a policy perspective, regulators and hospital boards should require cyber risk–integrated economic evaluations for major IT infrastructure investments. Embedding cybersecurity risk into financial decision-making supports improved transparency, accountability, and long-term sustainability of healthcare delivery systems [18], [13].

10.3 Limitations

Case estimates are hypothetical and the real-world data breach frequencies and costs will vary across facilities and the size of health care organizations.

11. CONCLUSION

This study presents a cyber risk–integrated cost–benefit analysis framework designed to support HIPAA-compliant IT infrastructure modernization in hospitals. By explicitly monetizing cybersecurity risk and embedding regulatory compliance into discounted cash-flow analysis, the framework enables healthcare administrators, CIOs, and compliance leaders to make informed, economically justified infrastructure investment decisions. The proposed approach addresses a critical gap in existing healthcare economic evaluations and demonstrates that cyber risk reduction can serve as a primary economic driver rather than a secondary consideration. Future research may extend this framework using real-world longitudinal data and comparative multi-hospital implementations.

12. ETHICAL AND LEGAL CONSIDERATIONS

This study is based on secondary data, industry reports, and hypothetical hospital scenarios. No identifiable patient data or real hospital IT logs were accessed. As such, institutional review board approval was not required. The framework is designed to support compliance with HIPAA and related data protection regulations. Future research may apply the proposed framework to real hospital modernization projects to validate empirical cost and risk assumptions across diverse healthcare environments.

REFERENCES

1. Nguyen, K. H. (2024). Cashing in: A cost–benefit analysis framework for digital hospitals. *BMC Health Services Research*, 24, 112.
2. Thusini, S. (2022). The development of the concept of return-on-investment in healthcare quality improvement: A systematic review. *International Journal for Quality in Health Care*, 34(2), mzac012.

3. Marques, I. C. P., Ferreira, J. J. M., & Ratten, V. (2023). Hospital costing methods: Four decades of literature. *Healthcare*, 11(3), 412.
4. Vassolo, R. S., Montoro-Sánchez, Á., & Lasserre, P. (2021). Hospital investment decisions in Health care 4.0 environments. *Technological Forecasting and Social Change*, 169, 120823.
5. Voets, M. M., et al. (2022). Economic evaluations of artificial intelligence applications in healthcare: A systematic review. *Health Economics Review*, 12(1), 38.
6. Anderson, R. (2001). Why information security is hard – An economic perspective. *IEEE Computer*, 34(4), 103–105.
7. Gordon, L. A., & Loeb, M. P. (2002). The economics of information security investment. *ACM Transactions on Information and System Security*, 5(4), 438–457.
8. Gordon, L. A., & Loeb, M. P. (2002). The economics of information security investment. *ACM Transactions on Information and System Security*, 5(4), 438–457.
<https://doi.org/10.1145/581271.581274>
9. Gordon, L. A., Loeb, M. P., Lucyshyn, W., & Zhou, L. (2015). Externalities and the magnitude of cybersecurity underinvestment by private sector firms. *Journal of Information Security*, 6(1), 24–31. <https://doi.org/10.4236/jis.2015.61003>
10. Menachemi, N., Chukmaitov, A., Saunders, C., & Brooks, R. G. (2018). Hospital adoption of information technologies and improved patient safety: A study of U.S. acute care hospitals. *Journal of the American Medical Informatics Association*, 25(1), 25–33.
<https://doi.org/10.1093/jamia/ocx050>
11. Kruse, C. S., Frederick, B., Jacobson, T., & Monticone, D. K. (2017). Cybersecurity in healthcare: A systematic review of modern threats and trends. *Technology and Health Care*, 25(1), 1–10. <https://doi.org/10.3233/THC-161263>
12. Ronquillo, J. G., Erik Winterholler, J., Cwikla, K., Szymanski, R., & Levy, C. (2018). Readiness of hospitals to manage cybersecurity threats: A national survey. *Journal of Medical Internet Research*, 20(10), e292. <https://doi.org/10.2196/jmir.9761>
13. Adler-Milstein, J., Ronchi, E., Cohen, G. R., Winn, L. A., & Jha, A. K. (2014). Benchmarking health IT among OECD countries. *Health Affairs*, 33(4), 698–706.
<https://doi.org/10.1377/hlthaff.2013.1323>
14. Dranove, D., Forman, C., Goldfarb, A., & Greenstein, S. (2015). The trillion-dollar conundrum: Complementarities and health information technology. *American Economic Journal: Economic Policy*, 7(4), 239–270. <https://doi.org/10.1257/pol.20120281>
15. Lluch, M. (2011). Healthcare professionals’ organisational barriers to health information technologies—A literature review. *International Journal of Medical Informatics*, 80(12), 849–862. <https://doi.org/10.1016/j.ijmedinf.2011.09.005>
16. Aceto, G., Persico, V., & Pescapé, A. (2018). Industry 4.0 and health: Internet of Things, big data, and cloud computing for healthcare 4.0. *Journal of Industrial Information Integration*, 18, 100129. <https://doi.org/10.1016/j.jii.2020.100129>
17. Raghupathi, W., & Raghupathi, V. (2014). Big data analytics in healthcare: Promise and potential. *Health Information Science and Systems*, 2(1), 3. <https://doi.org/10.1186/2047-2501-2-3>

18. Weill, P., & Ross, J. W. (2004). IT governance: How top performers manage IT decision rights for superior results. *MIS Quarterly Executive*, 3(4), 191–203.
<https://doi.org/10.2139/ssrn.664612>
19. Kwon, J., Johnson, M. E., & Sohn, S. G. (2013). Return on security investment (ROSI): A practical framework. *Journal of Cybersecurity*, 1(1), 45–52.
<https://doi.org/10.1093/cybsec/tyu003>

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